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Network Flow Problems as LPs

Review Learning Objectives

Submit your assignment

Due Feb 11, 11:59 PM IST

Receive grade

To Pass 80% or higher

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Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

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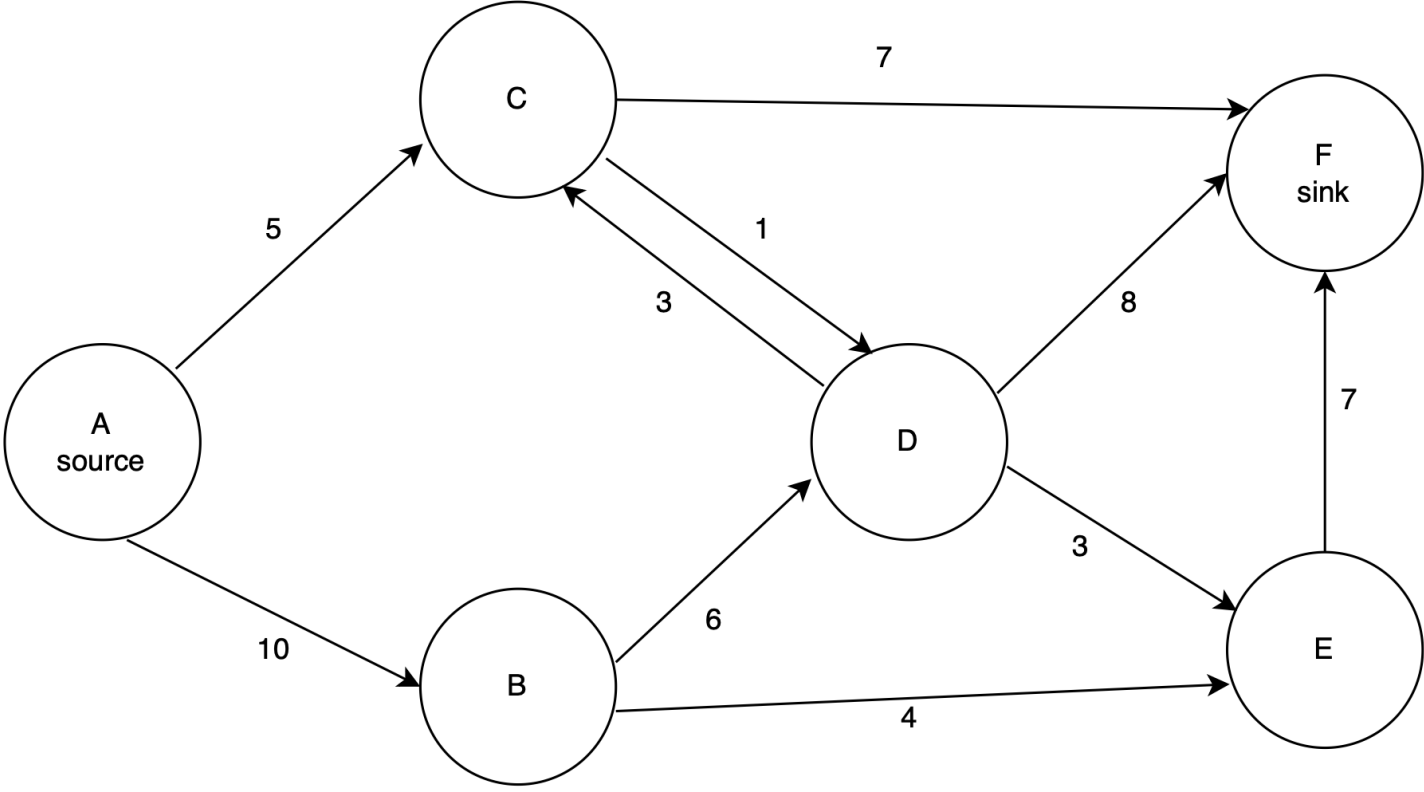
1. Select all the options that are true for network flow problems. Assume all edge capacities are finite and non-negative.2 / 2 points
- ☒ At each vertex , the incoming flow equals the outgoing flow.

☒ Correct
Correct: flow conservation is followed
- ☐ There should not exist any cycles in the network.

☒ Flow capacity can be less than or equal to the capacity of the edge.
- ☒ Correct
Correct: Flow along an edge can be less than or equal to the capacity of the edge , but cannot exceed the capacity.
- ☐ Network flow problems may be unbounded.

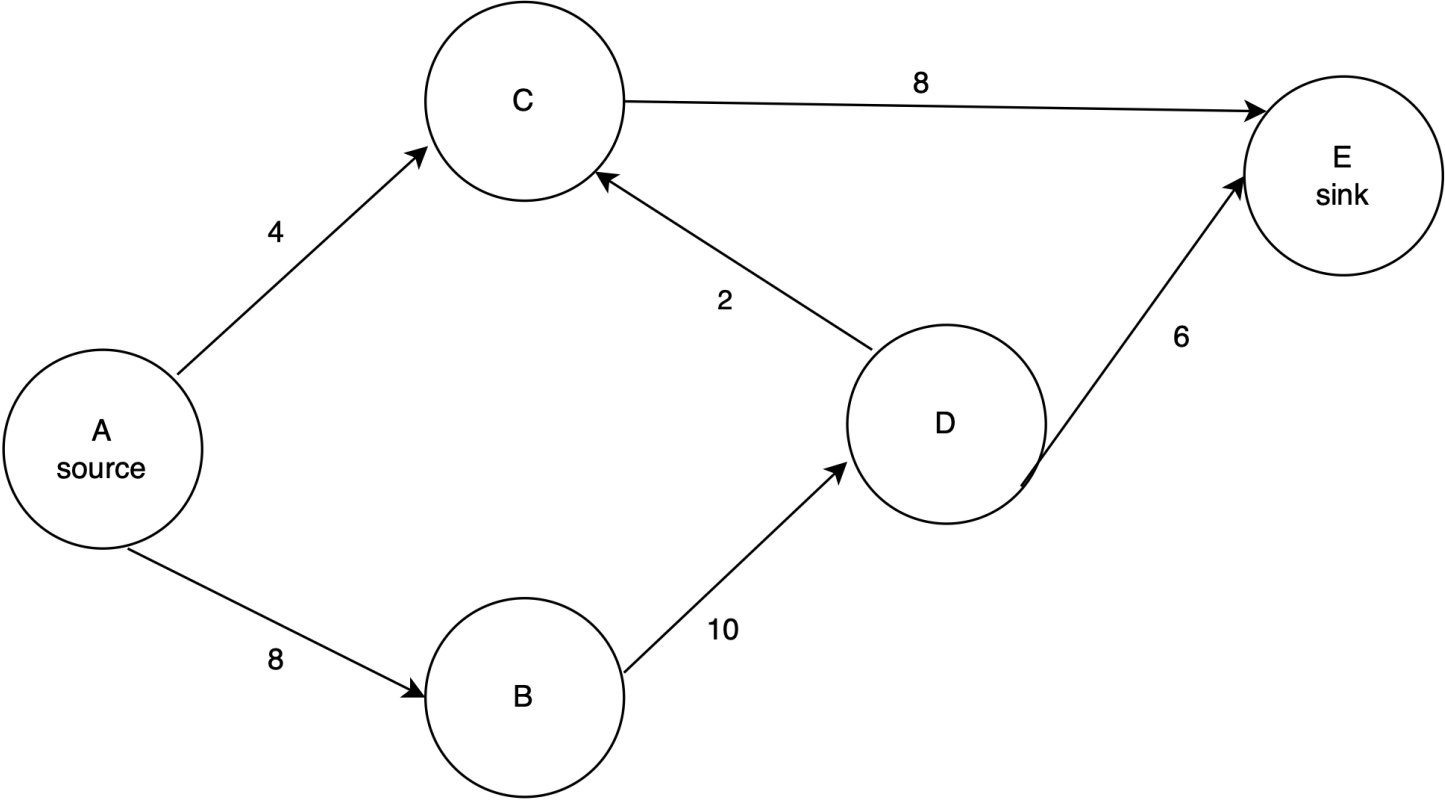
☒ Network flow problems are always feasible as long as the graph is connected.
- ☒ Correct
Correct. Since we can always send 0 flow through all the edges and that satisfies all the constraints.

2. For the given network , what is the maximum flow that can be transported from source to sink ?1 / 1 point



- ☒ 15
- ☐ 22
- ☐ 18
- ☐ 20
- ☒ Correct

- 3.5 / 5 points



Select all the correct flow conservation constraints from the list below.

- ☒

$x_{BD} = x_{DC} + x_{DE}$
- ☒ Correct
Correct: This is the conservation constraint for vertex D.
- ☐

$x_{BD} + x_{CD} = x_{DE}$
- ☐

$x_{AC} + x_{AB} = 0$
- ☒ $x_{AB} = x_{BD}$

☒ Correct
Correct : this is the constraint for vertex B.
- ☒ $x_{AC} + x_{DC} = x_{CE}$

☒ Correct
Correct one for vertex C

